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Direct Allylic Amination of Allylic Alcohol Catalyzed by Palladium Complex Bearing Phosphine−Borane Ligand

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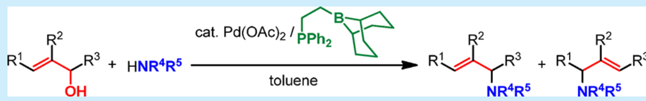
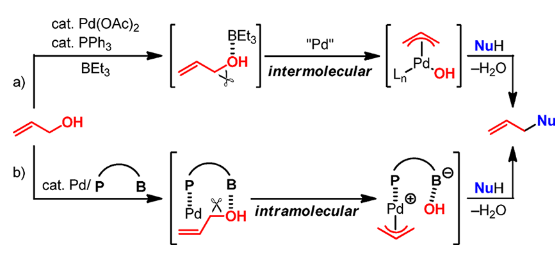
Supporting Information

ABSTRACT: The direct electrophilic, nucleophilic, and amphiphilic allylations of allylic alcohol by use of a palladium catalyst and organometallic reagents such as organoborane and organozinc has been developed. The phosphine−borane compound works as the effective ligand for palladium-catalyzed direct allylic amination of allylic alcohol. Thus, with secondary amines, the reaction was completed in only 1 h, even at room temperature.



Catalytic allylic substitution starting from the Tsuji−Trost reaction[1] is one of the most useful processes for organic synthesis. In the original Tsuji−Trost reaction, various allylic electrophiles, such as allylic chlorides, acetates, carbonates, and other types of allylic esters, could be used to give the allylated products. These allylic electrophiles could be derived from allyl alcohols via halogenation and esterification accompanied by the formation of a stoichiometric amount of byproduct. From the viewpoint of the step-economy synthesis and the reduction of waste, the development of highly efficient direct allylic substitution of allyl alcohols is in demand in the field of organic synthesis.[2,3] The direct allylic substitution gives the corresponding allylic compound in one step along with only 1 equiv of water as a byproduct. In our laboratory, we have been studying the direct allylic substitution by use of a palladium catalyst and organometallic reagents such as organoborane and organozinc. In the course of our study, we have so far developed the direct electrophilic allylation[4] as well as the direct nucleophilic[4i,5] and amphiphilic[4i,6] allylation using allyl alcohols. The electrophilic direct allylic substitution catalyzed by a palladium species and BEt_3 is shown in Scheme 1 (a). In this reaction, allylic alcohol was activated by Lewis acidic BEt_3 to give π-allylpalladium species via an oxidative addition. This π-allylpalladium was an electrophilic species and attacked by a nucleophile to give the product and water. We think that the rate-determining step of this reaction is an oxidative addition of allyl alcohol cleaving the C−O bond. We have now thought that the modification of this intermolecular oxidative addition to intramolecular process should be effective for the acceleration of the direct allylic substitution. To execute the intramolecular oxidative addition, we designed a palladium catalyst bearing a Lewis acid moiety. The phosphine−borane compound was expected to be a suitable ligand for this idea. Our working hypothesis is shown in Scheme 1 (b). Boryl group of the Pd/phosphine−borane catalyst acts as Lewis acid to activate the hydroxy group of allyl alcohols. Successive oxidative addition proceeds intramolecularly to give the π-allylpalladium intermediate which can be attacked by a nucleophile.

Scheme 1. Plausible Reaction Pathway for the Direct Allylic Substitution[a]



[a]Pathway b is our working hypothesis.

The synthesis, structure analysis, and reactivity of various transition-metal complexes bearing monophosphine−boranes[7] and monophosphine−borates[7e,8] as ligands have been reported by many groups. Some of these phosphine−borane complexes are useful for the catalytic reaction.[9,10] Pd-catalyzed Suzuki−Miyaura coupling[11] and Rh-catalyzed hydroformylation[12] using phosphine−borane ligands were reported by Bourissou and coworkers. In their initial works, o-(dimesitylboryl)phenylphosphine ligand behaved like a biaryl phosphine ligand. The weak π-coordination of the mesityl group linked to boron to the metal center was shown by X-ray crystal structure analysis, and this coordination enhanced the catalytic activity. On the other hand, we have tried to use the Lewis acidity of the borane moiety of phosphine−borane ligands to catalyze the direct allylic substitution. Herein, we report the palladium-catalyzed direct allylic amination of allyl alcohols by using phosphine−boranes as ligands.

Palladium-catalyzed allylic amination of cinnamyl alcohol (**1a**) with N-methylaniline (**2a**) was investigated using various types of phosphine−borane and phosphine−boronate ligands in tetrahy-

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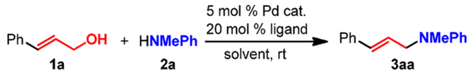
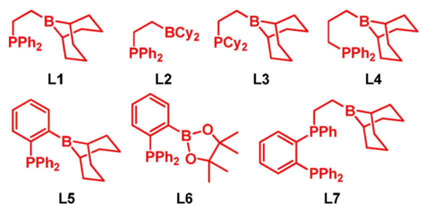
Table 1. Reactions of Cinnamyl Alcohol (1a) with *N*-Methylaniline (2a)[a]



entry	Pd catalyst	ligand	solvent	time (h)	yield[b] (%)
1	Pd(OAc)$_2$	PPh$_3$[c]	THF	24	99
2	Pd(OAc)$_2$	**L1**	THF	24	91
3	Pd(OAc)$_2$	**L2**	THF	24	2
4	Pd(OAc)$_2$	**L3**	THF	24	0
5	Pd(OAc)$_2$	**L4**	THF	24	64
6	Pd(OAc)$_2$	**L5**	THF	24	33
7	Pd(OAc)$_2$	**L6**	THF	24	2
8	Pd(OAc)$_2$	**L7**[d]	THF	24	0
9	PdCl$_2$	**L1**	THF	24	0
10	Pd$_2$(dba)$_3$[e]	**L1**	THF	24	2
11	Pd(OAc)$_2$	**L1**	toluene	3	99
12	Pd(OAc)$_2$[e]	**L1**[d]	toluene[f]	1	98
13	Pd(OAc)$_2$[e]	**L1**[g]	toluene[f]	1	99
14	Pd(OAc)$_2$[e]	**L1**[h]	toluene[f]	1.5	89
15	Pd(OAc)$_2$[e]	**L1**[i]	toluene[f]	1.5	5

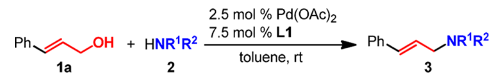
[a]A mixture of **1a** (1.2 mmol), **2a** (1.0 mmol), Pd catalyst, ligand, and solvent (5 mL) was stirred at rt under N$_2$. [b]Isolated yield. [c]30 mol % of BEt$_3$ was added. [d]10 mol % of ligand was used. [e]2.5 mol % of Pd catalyst was used. [f]1 mL of toluene was used. [g]7.5 mol % of **L1** was used. [h]5 mol % of **L1** was used. [i]2.5 mol % of **L1** was used.



drofuran (THF) as solvent (Table 1). The complete regioselectivity was observed in all cases to give the linear allylamine product **3aa**. The best ligand for this reaction was revealed to be Ph$_2$PCH$_2$CH$_2$(9-BBN) (**L1**) (entry 2). The yield was the same level as the reaction using Pd/PPh$_3$/BEt$_3$ catalyst system (entry 1). The ligands having dicyclohexylboryl group and dicyclohexylphosphino group (**L2** and **L3**) did not show the catalytic activity (entries 3 and 4), showing that the combination of the compound on phosphine and boron is important in this reaction. Propylene- and *o*-phenylene-linked phosphine−borane ligands (**L4** and **L5**) were less effective than **L1** (entries 5 and 6). Almost no reaction was observed by use of phosphine-boronate ligand **L6** bearing a pinacol unit (entry 7). These results show that the length and the moderate flexibility of ethylene linker in **L1** are well matched to this reaction. As bidentate diphosphines seemed to be the useful ligands for the catalysis, we newly prepared the diphosphine-borane **L7** and used this ligand for the direct allylic amination. However, unfortunately, the reaction did not proceed at all (entry 8). In place of Pd(OAc)$_2$, PdCl$_2$ and Pd$_2$(dba)$_3$ could not be used as a catalyst precursor in this reaction (entries 9 and 10). We think that the Pd(0) did not form from PdCl$_2$ under these reaction conditions, and ligand exchange did not proceed smoothly in the case of Pd$_2$(dba)$_3$. By use of toluene as solvent instead of THF, the reaction catalyzed by Pd/**L1** was completed for 3 h to give allylated product in quantitative yield (entry 11). Catalyst loading could be reduced under more concentrated conditions (entry 12). In the case of the Pd(OAc)$_2$/PPh$_3$/BEt$_3$ catalyst system, longer reaction time was needed, and the reaction was completed after 24 h. This result shows that the allylic amination was accelerated by using the Pd/**L1** catalyst system. As to the ratio of **L1** to Pd, 3 or 2 equiv of **L1** to Pd was enough (entries 13 and 14), but an equimolar amount of **L1** to Pd was not effective for this reaction (entry 15). These results suggest that 1 equiv of **L1** is used as a reductant for Pd(OAc)$_2$ to form Pd(0) species, and another **L1** coordinates to Pd to form a 1:1 complex as a catalyst molecule.

Reactions of cinnamyl alcohol (**1a**) with various secondary amines **2b−m** under the optimized reaction conditions (Table 1, entry 13) are summarized in Table 2. The regioselectivity in all

Table 2. Reactions of Cinnamyl Alcohol (1a) with Various Secondary Amines 2[a]



entry	2	temp (°C)	time (h)	3	yield[b] (%)
1	HNEtPh (**2b**)	rt	1.5	**3ab**	99
2	HNBnPh (**2c**)	rt	1	**3ac**	99[c]
3	HNBn$_2$ (**2d**)	rt	1.5	**3ad**	99
4[d]	HNPh$_2$ (**2e**)	80	1.5	**3ae**	80[e]
5	HN^{i}Pr$_2$ (**2f**)	80	6	**3af**	86
6	HNCy$_2$ (**2g**)	80	1	**3ag**	83
7[f]	pyrrolidine (**2h**)	80	12	**3ah**	98
8	piperidine (**2i**)	80	12	**3ai**	99
9	morpholine (**2j**)	80	1	**3aj**	91
10	*N*-phenyl-piperazine (**2k**)	rt	12	**3ak**	90
11	indoline (**2l**)	rt	3	**3al**	99
12	dibenzazepine (**2m**)	rt	1	**3am**	97[g]

[a]A mixture of **1a** (1.2 mmol), **2** (1.0 mmol), Pd(OAc)$_2$ (0.025 mmol), **L1** (0.075 mmol), and toluene (1 mL) was stirred under N$_2$. [b]Isolated yield. [c]**3ac** was obtained as a mixture of *E*/*Z* (85/15). [d]The reaction of **1a** with **2e** in the presence of Pd(OAc)$_2$ (2.5 mol %), PPh$_3$ (7.5 mol %), and BEt$_3$ (7.5 mol %) gave **3ae** (*E*/*Z* = 97/3) in 31% yield after 1.5 h. [e]**3ae** was obtained as a mixture of *E*/*Z* (93/7). [f]The reaction of **1a** with **2h** in the presence of Pd(OAc)$_2$ (2.5 mol %), PPh$_3$ (7.5 mol %), and BEt$_3$ (7.5 mol %) gave no product after 12 h. [g]**3am** was obtained as a mixture of *E*/*Z* (85/15).

reactions was perfect to give the corresponding cinnamyl amine **3ab−am** as a single product. *N*-Ethylaniline (**2b**), *N*-benzylaniline (**2c**), and dibenzylamine (**2d**) gave the products **3ab**, **3ac**, and **3ad** in quantitative yields (entries 1−3). The higher reaction temperature was needed for sterically more hindered amines **2e−g** (entries 4−6). The reactions of **1a** with various cyclic secondary amines **2h−m** afforded the corresponding amines **3ah−am** in high yields (entries 7−12). For the purpose of comparison of the phosphine−borane ligand with external BEt$_3$, two reactions of **1a** with **2e** and **2h** in the presence of Pd(OAc)$_2$, PPh$_3$, and BEt$_3$ were investigated. The corresponding amines **3ae** and **3ah** were obtained in lower yield (footnotes [d] and [f]).

As an example of the reaction using a primary amine, the reaction of cinnamyl alcohol (**1a**) with aniline (**2n**) was investigated (Scheme 2). The reaction proceeded at room

Scheme 2. Reaction of Cinnamyl Alcohol (1a) with Aniline (2n)

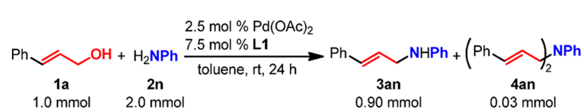


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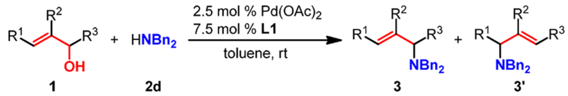
Table 3. Reactions of Various Allyl Alcohols 1 with Dibenzylamine (2d)[a]



entry	R^1	R^2	R^3	1	time (h)	products	yield (%)[b]	3/3′[c]	E/Z[d]
1[e]	H	H	H	**1b**	0.5	**3bd**	99		
2	4-ClC_6H_4	H	H	**1c**	1.5	**3cd** and **3′cd**	88	>99/1	>99/1
3	2-MeC_6H_4	H	H	**1d**	1	**3dd** and **3′dd**	80	>99/1	>99/1
4	3-MeC_6H_4	H	H	**1e**	1	**3ed** and **3′ed**	95	>99/1	>99/1
5	4-MeC_6H_4	H	H	**1f**	1	**3fd** and **3′fd**	99	>99/1	>99/1
6	4-$MeOC_6H_4$	H	H	**1g**	1	**3gd** and **3′gd**	99	>99/1	>99/1
7	1-naphthyl	H	H	**1h**	1	**3hd** and **3′hd**	95	>99/1	>99/1
8	2-naphthyl	H	H	**1i**	1	**3id** and **3′id**	76	>99/1	>99/1
9	2-furyl	H	H	**1j**	1	**3jd** and **3′jd**	95	>99/1	>99/1
10	2-thienyl	H	H	**1k**	1	**3kd** and **3′kd**	99	>99/1	>99/1
11[f]	Me	H	H	**1l**	1.5	**3ld** and **3′ld**	97	76/24	88/12
12[g]	H	H	Ph	**1m**	0.5	**3′ad** and **3ad**	99	1/>99	>99/1
13	H	H	Me	**1n**	3	**3′ld** and **3ld**	97	18/82	93/7
14	H	Ph	H	**1o**	1.5	**3od**	87		
15	H	Me	H	**1p**	12	**3pd**	91		
16	Ph	Me	H	**1q**	3	**3qd** and **3′qd**	88	>99/1	88/12

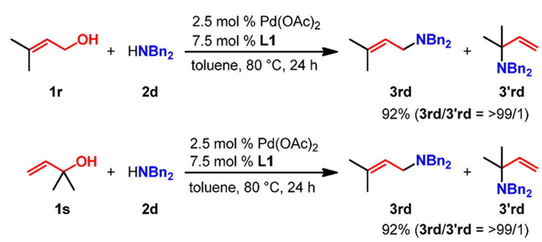
[a]A mixture of **1** (1.2 mmol), **2d** (1.0 mmol), $Pd(OAc)_2$, **L1**, and toluene was stirred at rt under N_2. [b]Isolated yield. [c]Determined by ^1H NMR. [d]Determined by ^1H NMR. [e]The reaction of **1b** with **2d** in the presence of $Pd(OAc)_2$ (2.5 mol %), PPh_3 (7.5 mol %), and BEt_3 (7.5 mol %) gave **3bd** in 18% yield after 0.5 h. [f]*E*/*Z* mixture of **1l** (*E*/*Z* = 83/17) was used. [g]The reaction of **1m** with **2d** in the presence of $Pd(OAc)_2$ (2.5 mol %), PPh_3 (7.5 mol %), and BEt_3 (7.5 mol %) gave **3ad** in 22% yield after 0.5 h.

temperature to give *N*-cinnamylaniline (**3an**) in 90% yield along with a small amount of *N*,*N*-dicinnamylaniline (**4an**) by use of 2 equiv of **2n**. Although we also investigated the reaction of **1a** with *n*-butylamine, the complex mixture including a small amount of *n*-butyl-bis(3-phenyl-2-propenyl)amine was obtained.

Various allylic alcohols could be used in this allylic amination. The results of the reactions with dibenzylamine (**2d**) are summarized in Table 3. Allyl alcohol (**1b**) gave the corresponding allylamine **3bd** quantitatively (entry 1). The reactions of cinnamyl alcohol derivatives **1c−g** proceeded to give the allylamines **3cd−gd** in high yields with almost complete regio- and stereoselectivities (entries 2−6). Some other 3-aryl- and 3-heteroaryl-2-propen-1-ols **1h−k** also reacted with **2d** to give the corresponding products **3hd−kd** in high yields with complete selectivities (entries 7−10). When crotyl alcohol (**1l**) was used, the excellent yield of the mixture of allylamines **3ld** and **3′ld** was obtained in a ratio of 76/24 (entry 11). Cinnamylamine **3ad** was obtained by the reaction of 1-phenyl-2-propen-1-ol (**1m**) with **2d** in a quantitative yield as a single product (entry 12). 3-Buten-2-ol (**1n**) gave the products as a mixture of **3ld** and **3′ld** in 97% yield (entry 13). The yields and the ratios of products from α-substituted allylic alcohols **1m** and **1n** were nearly the same as in the cases of cinnamyl alcohol (**1a**) and crotyl alcohol (**1l**), suggesting that these allylation reactions proceeded via π-allylpalladium intermediates. β-Substituted allylic alcohols **1o** and **1p** could also be used in this reaction to give the corresponding allylamines in high yields, respectively (entries 14 and 15). 2-Methyl-3-phenyl-2-propen-1-ol (**1q**) gave the product **3qd** in high yield with a complete regioselectivity, but the *E*-selectivity was slightly lower (entry 16). We investigated the reactions of **1b** and **1m** by using a $Pd(OAc)_2/PPh_3/BEt_3$ catalyst system to show the beneficial influence of the ligand **L1**. The lower yields of **3bd** and **3ad** were observed (footnotes [e] and [g]).

The remarkable effect of linking the phosphine and borane moiety was observed in the amination reaction of prenyl alcohol (**1r**) and 2-methyl-3-buten-2-ol (**1s**). By use of the $Pd(OAc)_2$/**L1** catalyst system, prenylamine **3rd** was obtained in a high yield with a complete regioselectivity from both **1r** and **1s** (Scheme 3).

Scheme 3. Reactions of Prenyl Alcohol (1r) and 2-Methyl-3-buten-2-ol (1s) with Dibenzylamine (2d)



On the other hand, when $EtPPh_2$ and *B*-*n*-hexyl-9-BBN were used instead of **L1**, the yield of the desired products **3rd** and **3′rd** was extremely low and the complex mixture was formed. In the case of the $Pd/PPh_3/BEt_3$ system, almost no reaction was observed after 24 h. These results indicate that the linkage between phosphine and borane moieties is critical for obtaining the high catalytic activity. We think that the activation of the allylic C−O bond and successive intramolecular oxidative addition are achieved by the catalysis of phosphine-borane palladium complex as shown in Scheme 1 (b).

In summary, we have now found that the phosphine−borane compound can be used as an effective ligand in the palladium-catalyzed direct allylic amination of allylic alcohol. The structure of the linker between phosphine and borane moieties in the ligand is important for obtaining high catalytic activity. Further studies on expanding the scope of nucleophiles are currently in progress.

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■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.orglett.7b03023.

Experimental procedures and detailed characterization data for the compounds (PDF)

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Notes

The authors declare no competing financial interest.

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